

GOVERNMENT TECHNICAL INSTITUTE (
pyinoolwin)

Department of Information Technology

IT-DP 2506 (Object Oriented Programming in
C++)

JOB SHEET

2025- Academic Year

A.G.T.I Second year (SemII)

Submitted by

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Job 1

Implementation of C++ Programming Code Introduction to C++ Program Code

1. Write a C++ program to print multiplication table of a given number.

Description:

For example, for a number 2 with 3 rows, the output should be:

$2 * 1 = 2$

$2 * 2 = 4$

$2 * 3 = 6$

At the time of execution, the program should print the following messages one by one

on the console as:

Enter an integer number :

Enter number of rows :

For example, if the user gives the input as:

```
Enter number u want table : 5
Enter upto how many numbers u want : 4
Multiplication table of 5 is :
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
int number, rows;
```

```
cout << "Enter number you want table: ";
```

```
cin >> number;
```

```
cout << "Enter upto how many numbers you want: ";
```

```
cin >> rows;
```

```
cout << "Multiplication table of " << number << " is:" << endl;
```

```
for (int i = 1; i <= rows; i++)
```

```
{
```

```
cout << number << " * " << i << " = " << number * i << endl;
```

```
}
```

```
return 0; }
```

```
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ASUS>gcc --version
gcc (x86_64-win32-sjlj-rev0, Built by MinGW-W64 project) 8.1.0
Copyright (C) 2018 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

C:\Users\ASUS>cd/

C:\>cd C++

C:\C++>cd C++/lab

C:\C++\Lab>g++ Lab1.1.cpp -o Lab1.1

C:\C++\Lab>Lab1.1
Enter number you want table: 5
Enter upto how many numbers you want: 4
Multiplication table of 5 is:
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
```

2. Write a CPP program to find prime numbers up to n values (n must be user defined).

Description:

At the time of execution, the program should print the message on the console as:

Enter the value of N to find prime numbers up to:

For example, if the user gives the input as:

```
Enter the value of N to find prime numbers upto : 20
2 3 5 7 11 13 17 19
```

Solution

```
#include<iostream>

using namespace std;

int main() {

    int n;

    cout << "Enter the value of N to find prime numbers up to: ";

    cin >> n;

    for (int i = 2; i <= n; i++) {

        bool Prime = true;

        for (int j = 2; j * j <= i; j++) {

            if (i % j == 0) {

                Prime = false;

                break;}

        }

        if (Prime) {

            cout << i << " ";}

    }

    cout << endl;
```

```
    return 0;
}
```

```
C:\C++\Lab>g++ Lab1.2.cpp -o Lab1.2
C:\C++\Lab>Lab1.2
Enter the value of N to find prime numbers up to: 34
2 3 5 7 11 13 17 19 23 29 31
```

3. Write a C++ program to perform arithmetic operations on two numbers. (Use switch case).

Description:

At the time of execution, the program should print the message on the console as:

Enter two integer values :

Enter an arithmetic operator :

For example, if the user gives the input as:

```
Enter a and b values : 12 10
CatLog To Select Certain Operations
+ -> Addition
- -> Subtraction
* -> Multiplication
/ -> Division
% -> Modulo Division
Enter any above sign to perform certain operations : +
Addition of 12and 10 is : 22
```

For example, if the input given as

```
Enter a and b values : 12 6
CatLog To Select Certain Operations
+ -> Addition
- -> Subtraction
* -> Multiplication
/ -> Division
% -> Modulo Division
Enter any above sign to perform certain operations : /
Division of 12and 6 is : 2
```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a, b;
```

```
    char option;
```

```
    cout << "Enter two integer values: ";
```

```
    cin >> a >> b;
```

```
    cout << "CatLog To Select Certain Operations\n";
```

```
    cout << "+ -> Addition\n";
```

```
    cout << "- -> Subtraction\n";
```

```
    cout << "* -> Multiplication\n";
```

```
    cout << "/ -> Division\n";
```

```
cout << "% -> Modulo Division\n";
```

```
cout << "Enter any above sign to perform certain operations: ";
```

```
cin >> option;
```

```
switch(option) {
```

```
    case '+':
```

```
        cout << "Addition of " << a << " and " << b << " is: " << (a + b) << endl;
```

```
        break;
```

```
    case '-':
```

```
        cout << "Subtraction of " << a << " and " << b << " is: " << (a - b) << endl;
```

```
        break;
```

```
    case '*':
```

```
        cout << "Multiplication of " << a << " and " << b << " is: " << (a * b) << endl;
```

```
        break;
```

```
    case '/':
```

```
        if (b != 0)
```

```
            cout << "Division of " << a << " and " << b << " is: " << (a / b) << endl;
```



```
else

    cout << "Error: Division by zero is not allowed." << endl;

    break;

case '%':

    if (b != 0)

        cout << "Modulo Division of " << a << " and " << b << " is: " << (a %
b) << endl;

    else

        cout << "Error: Modulo by zero is not allowed." << endl;

    break;

default:

    cout << "Invalid operator!" << endl; }

return 0;

}
```

```
C:\C++\Lab>g++ Lab1.3.cpp -o Lab1.3
C:\C++\Lab>Lab1.3
Enter two integer values: 4 5
CatLog To Select Certain Operations
+ -> Addition
- -> Subtraction
* -> Multiplication
/ -> Division
% -> Modulo Division
Enter any above sign to perform certain operations: +
Addition of 4 and 5 is: 9
```

Job 2

Implementation of C++ Programming Code using With Function Operation

1. Write a C++ program to swap two numbers using:

- a. Call -by-value
- b. Call -by-reference

Description:

Call -by-value

During execution, the program should print the following message on the console as:

```
Enter a and b values : 22 44
Swap two numbers by call by value
After swapping by call by value :
a = 44
b = 22
Swap two numbers by call by reference
After swapping by call by value :
a = 44
b = 22
```

Enter two integer values :

For example, if the user gives the following input as:

Call -by-reference

At the time of execution, the program should print the message on the console as:

Enter two integer values :

For example, if the user gives the input as:

Enter two integer values : 12 13

then the program should print the result as:

Before swapping in main : a = 12 b = 13

After swapping in swap : *p = 13 *q = 12

After swapping in main : a = 13 b = 12

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
void swapByValue(int x, int y) {
```

```
    int temp = x;
```

```
    x = y;
```

```
    y = temp;
```

```
    cout << "After swapping by call by value :" << endl;
```

```
    cout << "a = " << x << endl;
```

```
    cout << "b = " << y << endl;}
```

```
void swapByReference(int &x, int &y) {
```

```
    int temp = x;
```

```
    x = y;
```

```
    y = temp;
```

```
    cout << "After swapping by call by reference :" << endl;
```

```
    cout << "a = " << x << endl;
```

```
    cout << "b = " << y << endl;}
```

```
int main() {
```

```
    int a, b;
```

```
    cout << "Enter a and b values : ";
```

```
cin >> a >> b;

cout<<"Before swapping in main:"<<endl<<a<<endl<<b;

cout << "\nSwap two numbers by call by value" << endl;

swapByValue(a, b);

cout << "In main after call by value :" << endl;

cout << "a = " << a << endl;

cout << "b = " << b << endl;

cout << "Swap two numbers by call by reference" << endl;

swapByReference(a, b);

cout << "In main after call by reference :" << endl;

cout << "a = " << a << endl;

cout << "b = " << b << endl;

return 0;

}
```

```

C:\C++\Lab>Lab2.1
Enter a and b values : 3 4
Before swapping in main:
3
4
Swap two numbers by call by value
After swapping by call by value :
a = 4
b = 3
In main after call by value :
a = 3
b = 4
Swap two numbers by call by reference
After swapping by call by reference :
a = 4
b = 3
In main after call by reference :
a = 4
b = 3

```

2. Write a C++ program to find factorial of a given number. (using recursive and non-recursive

functions)

Description:

At the time of execution, the program should print the message on the console as:

Enter an integer :

For example, if the user gives the input as:

```

Enter a number to find it's factorial: 5
Factorial without using recursive function
Factorial of given number 5 is 120
Factorial with using recursive function
Factorial of given number 5 is 120

```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
int factorialNonRecursive(int n) {
```

```
    int fact = 1;
```

```
    for (int i = 1; i <=n; i++) {
```

```
        fact *= i;  }
```

```
    return fact;} 
```

```
int factorialRecursive(int n) {
```

```
    if (n == 0 || n == 1)
```

```
        return 1;
```

```
    else
```

```
        return n * factorialRecursive(n - 1);}
```

```
int main() {
```

```
    int num;
```

```
    cout << "Enter a number to find it's factorial: ";
```

```
    cin >> num;
```

```
    cout << "Factorial without using recursive function" << endl;
```

```
    cout << "Factorial of given number " << num << " is "
```

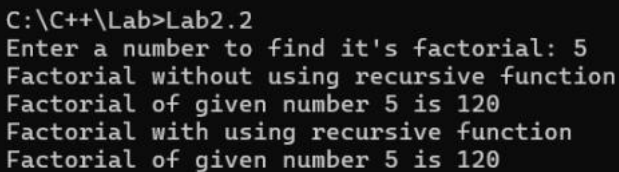
```
<< factorialNonRecursive(num) << endl; cout << "Factorial with using recursive function" << endl;
```

```
cout << "Factorial of given number " << num << " is "
```

```
<< factorialRecursive(num) << endl;
```

```
return 0;
```

```
}
```



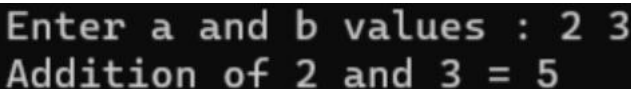
```
C:\C++\Lab>Lab2.2
Enter a number to find it's factorial: 5
Factorial without using recursive function
Factorial of given number 5 is 120
Factorial with using recursive function
Factorial of given number 5 is 120
```

3. Write a C++ program to find addition of two given numbers using inline function.

Description:

At the time of execution, the program should print the message on the console as:

Enter a and b values :For example, if the user gives the input as:



```
Enter a and b values : 2 3
Addition of 2 and 3 = 5
```

Solution

```
#include <iostream>
```



```
using namespace std;

int add(int x, int y) {

    return x + y;}

int main() {

    int a, b;

    cout << "Enter a and b values: ";

    cin >> a >> b;

    cout << "Addition of " << a << " and " << b << " = " << add(a, b) << endl;

    return 0;

}
```

```
C:\C++\Lab>Lab2.3
Enter a and b values: 5 6
Addition of 5 and 6 = 11
```

Job 3

Implementation of C++ Programming Code using With Object and Class

1. Write all the following programs using Object-Oriented implementation (using classes and

objects).

a. Write a C++ program to find sum and average of N numbers.

b. Write a C++ program to print squares of numbers and cubes of numbers between 1

and N.

c. Write a C++ program to print Fibonacci series up to N.

Description:

At the time of execution, the program should print the message on the console as:

Enter the number u want: 10

then the program should print the result as:

```
Enter the number u want : 10
Sum of first 10 numbers is = 55
Average of the 10 numbers is = 5

Enter the upto which number u want for sauares and cubes : 10
Squares of first 10 are
1 4 9 16 25 36 49 64 81 100
Cubes of first 10 are
1 8 27 64 125 216 343 512 729 1000
Fibonacci series
Enter the number of terms: 6
Fibonacci Series: 0 1 1 2 3 5
```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
class SumAverage {
```

```
    int n;
```

```
public:
```

```
    SumAverage(int num) {
```

```
        n = num; }
```

```
    void calculate() {
```

```
        int sum = 0;
```

```
        for (int i = 1; i <= n; i++) {
```

```
            sum += i;}
```

```
        cout << "Sum of first " << n << " numbers is = " << sum << endl;
```

```
        cout << "Average of the " << n << " numbers is = " << sum / n << endl;}
```

```
};
```

```
class SquareCube {
```

```
    int n;
```

public:

```
SquareCube(int num) {
```

```
    n = num; }
```

```
void display() {
```

```
    cout << "\nSquares of first " << n << " are" << endl;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        cout << i * i << " ";    }
```

```
    cout << endl;
```

```
    cout << "Cubes of first " << n << " are" << endl;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        cout << i * i * i << " "; }
```

```
    cout << endl; }
```

```
};
```

```
class Fibonacci {
```

```
    int n;
```

public:

```
Fibonacci(int num) {
```

```
    n = num;}
```

```

void generate() {

    cout << "\nFibonacci series" << endl;

    cout << "Enter the number of terms: " << n << endl;

    cout << "Fibonacci Series: ";

    int a = 0, b = 1, c;

    cout << a << " " << b << " ";

    for (int i = 2; i < n; i++) {

        c = a + b;

        cout << c << " ";

        a = b;

        b = c; }

    cout << endl; }

};

int main() {

    int n;

    cout << "Enter the number u want : ";

    cin >> n;

    SumAverage sa(n);

```

```
sa.calculate();
```

```
cout << "\nEnter the upto which number u want for squares and cubes :  
";
```

```
cin >> n;
```

```
SquareCube sc(n);
```

```
sc.display();
```

```
cout << "\nEnter the number of terms: ";
```

```
cin >> n;
```

```
Fibonacci fib(n);
```

```
fib.generate();
```

```
return 0;}
```

```
C:\C++\Lab>Lab3.1  
Enter the number u want : 10  
Sum of first 10 numbers is = 55  
Average of the 10 numbers is = 5  
  
Enter the upto which number u want for squares and cubes : 10  
  
Squares of first 10 are  
1 4 9 16 25 36 49 64 81 100  
Cubes of first 10 are  
1 8 27 64 125 216 343 512 729 1000  
  
Enter the number of terms: 6  
  
Fibonacci series  
Enter the number of terms: 6  
Fibonacci Series: 0 1 1 2 3 5
```

2. Write a C++ program to find sum of two times (using objects as arguments concept).

Description:

At the time of execution, the program should print the message on the console as:

Enter Time hours :

Enter Time minutes :

then the program should print the result as:

```
Enter Time hours : 3
Enter Time minutes : 55
Enter Time hours : 4
Enter Time minutes : 29
Total hours is : 8
Total minutes is : 24
```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
class Time {
```

```
    int hours, minutes;
```

```
public:
```

```
    void input() {
```

```
cout << "Enter Time hours : ";
```

```
cin >> hours;
```

```
cout << "Enter Time minutes : ";
```

```
cin >> minutes;}
```

```
void display() {
```

```
    cout << "Hours: " << hours << " Minutes: " << minutes << endl; }
```

```
void addTime(Time t1, Time t2) {
```

```
    minutes = t1.minutes + t2.minutes;
```

```
    hours = t1.hours + t2.hours;
```

```
    if (minutes >= 60) {
```

```
        hours += minutes / 60;
```

```
        minutes = minutes % 60; }
```

```
}
```

```
void showResult() {
```

```
    cout << "Total hours is : " << hours << endl;
```

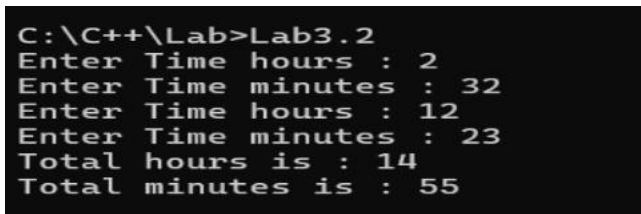
```
    cout << "Total minutes is : " << minutes << endl; }
```

```
};
```

```
int main() {
```



```
Time t1, t2, t3;  
  
t1.input();  
  
t2.input();  
  
t3.addTime(t1, t2);  
  
t3.showResult();  
  
return 0;  
  
}
```

A screenshot of a Windows command prompt window with a black background and white text. The prompt shows the directory 'C:\C++\Lab' and the file 'Lab3.2'. The program prompts the user to enter time in hours and minutes for two different times. The first time entered is 2 hours and 32 minutes. The second time entered is 12 hours and 23 minutes. The program then outputs the total hours as 14 and the total minutes as 55.

```
C:\C++\Lab>Lab3.2  
Enter Time hours : 2  
Enter Time minutes : 32  
Enter Time hours : 12  
Enter Time minutes : 23  
Total hours is : 14  
Total minutes is : 55
```

3. Write a C++ program to illustrate use of friend functions.

Description:

At the time of execution, the program should print the message on the console as:

Enter the real part :

Enter the imaginary part :

then the program should print the result as:

```
Enter the real part : 3
Enter the imaginary part : 6
Given Number is : 3 + 6i
Enter the real part : 1
Enter the imaginary part : 5
Given Number is : 1 + 5i
Given Number is : 4 + 11i
```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
class Complex {
```

```
    int real, imag;
```

```
public:
```

```
    void getData() {
```

```
        cout << "Enter the real part : ";
```

```
        cin >> real;
```

```
        cout << "Enter the imaginary part : ";
```

```
        cin >> imag;    }
```

```
    void showData() {
```

```
        cout << "Given Number is : " << real << " + " << imag << "i" << endl;    }
```

```
friend Complex add(Complex c1, Complex c2);
```

```
};
```

```
Complex add(Complex c1, Complex c2) {
```

```
    Complex temp;
```

```
    temp.real = c1.real + c2.real;
```

```
    temp.imag = c1.imag + c2.imag;
```

```
    return temp;}

```

```
int main() {
```

```
    Complex c1, c2, c3;
```

```
    c1.getData();
```

```
    c1.showData();
```

```
    c2.getData();
```

```
    c2.showData();
```

```
    c3 = add(c1, c2);
```

```
    c3.showData();
```

```
    return 0;
```

```
}
```

```
C:\C++\Lab>Lab3.3
Enter the real part : 3
Enter the imaginary part : 2
Given Number is : 3 + 2i
Enter the real part : 4
Enter the imaginary part : 3
Given Number is : 4 + 3i
Given Number is : 7 + 5i
```

Job 4

Implementation of C++ Programming Code using with Inheritance

1. Write a C++ program which implements single level inheritance

Description:

During execution, the program should print the following messages one by one on the

console as:

```
Enter Unique number of the student : 120
Enter the name of the student : Thiri
Enter height of the student : 157
Enter Weight of the student : 58

Name of the Student is : Thiri
Roll Number of the student is : 120
Height of the student is : 157
Weight of the student is : 58
```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
class Student {
```

```
public:
```

```
    int rollNo;
```

```
    string name;
```

```
    void getStudentData() {
```

```
        cout << "Enter Unique number of the student : ";
```

```
        cin >> rollNo;
```

```
        cout << "Enter the name of the student : ";
```

```
        cin >> name;  }
```

```
};
```

```
class PhysicalDetails : public Student {
```

```
public:
```

```
    int height;
```

```
    int weight;
```

```
    void getDetails() {
```

```
        cout << "Enter height of the student : ";
```

```
cin >> height;
```

```
cout << "Enter Weight of the student : ";
```

```
cin >> weight;}
```

```
void display() {
```

```
    cout << "\nName of the Student is : " << name << endl;
```

```
    cout << "Roll Number of the student is : " << rollNo << endl;
```

```
    cout << "Height of the student is : " << height << endl;
```

```
    cout << "Weight of the student is : " << weight << endl; }
```

```
};
```

```
int main() {
```

```
    PhysicalDetails obj;
```

```
    obj.getStudentData();
```

```
    obj.getDetails();
```

```
    obj.display();
```

```
    return 0;}
```

```
C:\C++\Lab>Lab4.1
Enter Unique number of the student : 120
Enter the name of the student : SHWE
Enter height of the student : 5
Enter Weight of the student : 119

Name of the Student is : SHWE
Roll Number of the student is : 120
Height of the student is : 5
Weight of the student is : 119
```

2. Write a C++ program which implements multi-level inheritance.

Description:

During execution, the program should print the following messages one by one on the

console as:

```
Enter Unique number of the student : 56
Enter the name of the student : Sandar

Name of the Student is : Sandar
Roll Number of the student is : 56
Enter student marks of subject 1 : 98
Enter student marks of subject 2 : 97
Enter student marks of subject 3 : 96

Marks of Subject -1 is = 98
Marks of Subject -2 is = 97
Marks of Subject -3 is = 96
Total = 291
Average = 97
```

Solution

```
#include <iostream>
```

```
using namespace std;
```

```
class Student {
```

public:

int rollNo;

string name;

void getStudentData() {

cout << "Enter Unique number of the student : ";

cin >> rollNo;

cout << "Enter the name of the student : ";

cin >> name;

}

void displayStudentData() {

cout << "\nName of the Student is : " << name << endl;

cout << "Roll Number of the student is : " << rollNo << endl;

}

};

class Marks : public Student {

public:

int m1, m2, m3;

void getMarks() {


```
    cout << "Enter student marks of subject 1 : ";

    cin >> m1;

    cout << "Enter student marks of subject 2 : ";

    cin >> m2;

    cout << "Enter student marks of subject 3 : ";

    cin >> m3;

}

void displayMarks() {

    cout << "Marks of Subject -1 is = " << m1 << endl;

    cout << "Marks of Subject -2 is = " << m2 << endl;

    cout << "Marks of Subject -3 is = " << m3 << endl;

}

};

class Result : public Marks {

public:

    void calculateResult() {

        int total = m1 + m2 + m3;

        float average = total / 3.0;
```

```
displayStudentData();
```

```
displayMarks();
```

```
cout << "Total = " << total << endl;
```

```
cout << "Average = " << average << endl;
```

```
}
```

```
};
```

```
int main() {
```

```
    Result obj;
```

```
    obj.getStudentData();
```

```
    obj.getMarks();
```

```
    obj.calculateResult();
```

```
    return 0;
```

```
}
```

```

C:\C++\Lab>Lab4.2
Enter Unique number of the student : 22
Enter the name of the student : Chaw
Enter student marks of subject 1 : 90
Enter student marks of subject 2 : 90
Enter student marks of subject 3 : 90

Name of the Student is : Chaw
Roll Number of the student is : 22
Marks of Subject -1 is = 90
Marks of Subject -2 is = 90
Marks of Subject -3 is = 90
Total = 270
Average = 90

```

3. Write a C++ program which implements hierarchical inheritance.

Description:

During execution, the program should print the following messages one by one on the

console as:

```

1.Savings account
2.Curent account
Enter any one option : 2
Enter acc_no of the user : 546731
Enter the name of the user : Sandar
Account number of the user is = 546731
Name of the user is : Sandar
Enter the balance amount in the users bank : 1800
The balance amount in your account is more than 1000 so need not to worry

```

```

1.Savings account
2.Curent account
Enter any one option : 1
Enter acc_no of the user : 21345
Enter the name of the user : Htwenge
Account number of the user is = 21345
Name of the user is : Htwenge
Enter the balance amount in the users bank : 2400
The balance amount in your account is more than 500 so need not to worry

```

Solution

```
#include <iostream>

using namespace std;

class Account {

public:

    long acc_no;

    string name;


    void getData() {

        cout << "Enter acc_no of the user : ";

        cin >> acc_no;

        cout << "Enter the name of the user : ";

        cin >> name;

    }

    void displayData() {

        cout << "Account number of the user is = " << acc_no << endl;

        cout << "Name of the user is : " << name << endl;

    }

};
```

```
class Savings : public Account {
```

```
public:
```

```
    float balance;
```

```
    void getBalance() {
```

```
        cout << "Enter the balance amount in the users bank : ";
```

```
        cin >> balance;
```

```
    }
```

```
    void checkBalance() {
```

```
        if(balance > 500)
```

```
            cout << "The balance amount in your account is more than 500 so  
need not to worry" << endl;
```

```
        else
```

```
            cout << "Balance is below minimum limit of 500!" << endl;
```

```
    }
```

```
};
```

```
class Current : public Account {
```

```
public:
```

```
    float balance;
```

```
void getBalance() {
```

```
    cout << "Enter the balance amount in the users bank : ";
```

```
    cin >> balance;
```

```
}
```

```
void checkBalance() {
```

```
    if(balance > 1000)
```

```
        cout << "The balance amount in your account is more than 1000 so  
need not to worry" << endl;
```

```
    else
```

```
        cout << "Balance is below minimum limit of 1000!" << endl;
```

```
}
```

```
};
```

```
int main() {
```

```
    int choice;
```

```
    cout << "1. Savings account" << endl;
```

```
    cout << "2. Current account" << endl;
```

```
    cout << "Enter any one option : ";
```

```
cin >> choice;

if(choice == 1) {

    Savings s;

    s.getData();

    s.displayData();

    s.getBalance();

    s.checkBalance(); }

else if(choice == 2) {

    Current c;

    c.getData();

    c.displayData();

    c.getBalance();

    c.checkBalance() }

else {

    cout << "Invalid option!" << endl; }

return 0}
```

```
C:\C++\Lab>Lab4.3
1. Savings account
2. Current account
Enter any one option : 1
Enter acc_no of the user : 5262
Enter the name of the user : Shwe
Account number of the user is = 5262
Name of the user is : Shwe
Enter the balance amount in the users bank : 5262
The balance amount in your account is more than 500 so need not to worry
```

Job 5

Implementation of C++ Programming Code using With Streams and File Creation

1.Write a program how to read an employee's details from keyboard using class and object then

write that object into the file? In this program there are following details to be read through

Employee class

Output


```
EMPLOYEE DETAILS
ENTER EMPLOYEE ID : 1001
ENTER NAME OF THE EMPLOYEE : Su Sandi
ENTER DESIGNATION : Student
ENTER DATE OF JOIN:
DATE : 21
MONTH: 12
YEAR : 2023
ENTER DATE OF BIRTH:
DATE : 15
MONTH: 09
YEAR : 2001
Date saved into file the file.
```

```
Data extracted from file..
EMPLOYEE ID: 1001
EMPLOYEE NAME: Su Sandi
DESIGNATION: Student
DATE OF JOIN: 21/12/2023
DATE OF BIRTH: 15/9/2001
```

- Employee ID

- Employee Name
- Designation
- Date of joining
- Date of birth

Solution

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
class Employee {
```

```
private:
```

```
    int emplID;
```

```
    string name;
```

```
    string designation;
```

```
    string doj;
```

```
    string dob;
```

```
public:
```

```
    void getData() {
```

```
        cout << "EMPLOYEE DETAILS\n";
```

```
        cout << "ENTER EMPLOYEE ID : ";
```

```
        cin >> emplID;
```

```
        cin.ignore();
```

```
cout << "ENTER NAME OF THE EMPLOYEE : ";

cin>>name;

cout << "ENTER DESIGNATION : ";

cin>>designation;

cout << "ENTER DATE OF JOIN (dd/mm/yyyy): ";

cin>>doj;

cout << "ENTER DATE OF BIRTH (dd/mm/yyyy): ";

cin>>dob;

}

void saveToFile() {

    ofstream MyFile("lab.txt");

    MyFile << emplID << endl;

    MyFile << name << endl;

    MyFile << designation << endl;

    MyFile << doj << endl;

    MyFile << dob << endl;

    MyFile.close();

    cout << "Data saved into file.\n";
```

```
}
```

```
void readFromFile() {
```

```
    string myText;
```

```
    ifstream MyReadFile("lab.txt");
```

```
    cout << "\nData extracted from file..\n";
```

```
    int line = 0;
```

```
    while (getline(MyReadFile, myText)) {
```

```
        if (line == 0) cout << "EMPLOYEE ID: " << myText << endl;
```

```
        else if (line == 1) cout << "EMPLOYEE NAME: " << myText << endl;
```

```
        else if (line == 2) cout << "DESIGNATION: " << myText << endl;
```

```
        else if (line == 3) cout << "DATE OF JOIN: " << myText << endl;
```

```
        else if (line == 4) cout << "DATE OF BIRTH: " << myText << endl;
```

```
        line++;
```

```
    }
```

```
    MyReadFile.close();
```

```
}
```

```
C:\C++\Lab>Lab5
EMPLOYEE DETAILS
ENTER EMPLOYEE ID : 234
ENTER NAME OF THE EMPLOYEE : Chaw
ENTER DESIGNATION : Student
ENTER DATE OF JOIN (dd/mm/yyyy): 18 5 2005
ENTER DATE OF BIRTH (dd/mm/yyyy): Data saved into file.

Data extracted from file..
EMPLOYEE ID: 234
EMPLOYEE NAME: Chaw
DESIGNATION: Student
DATE OF JOIN: 18
DATE OF BIRTH: 5
```

```
};
```

```
int main() {
```

```
    Employee e;
```

```
    e.getData();
```

```
    e.saveToFile();
```

```
    e.readFromFile();
```

```
    return 0;
```

```
}
```

Z234
Chaw
Student
18
5